Duke University
Math 218: Matrices and Vector Spaces

Exam 1 - Practice

These instructions will be on the test:

- Do not open this test booklet until you are directed to do so.
- You will have 50 minutes to complete the exam. If you finish early go back and check your work.
- This exam is closed book.
- You may use a calculator for arithmetic purpose only. No calculator matrix computations are allowed.
- Throughout the exam, show your work so that your reasoning is clear. Otherwise no credit will be given. Circle your answers.
- Do not spend too much time on any one problem. Read them all through first and attack them in an order that allows you to make the most progress.

Problem	Points	Grade
1	25	
2	5	
3	9	
4	6	
4	5	
Total	50	

Name:_____

1. (25 points) Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & 5 & 0 \\ 1 & 2 & 4 & 2 \\ 0 & -1 & 0 & 8 \\ -1 & -3 & -1 & -1 \end{pmatrix}.$$

The goal of this question is to solve the equation $A\vec{x} = [12, 1, -30, 6]^T$.

(a) Carry out Gaussian reduction with maximal partial pivoting to show that the upper triangular component of the $PA = LU$ decomposition is

$$U = \begin{pmatrix} 1 & 2 & 5 & 0 \\ 0 & -1 & 0 & 8 \\ 0 & 0 & 4 & -9 \\ 0 & 0 & 0 & -\frac{1}{4} \end{pmatrix},$$

and find the L and P components. You may want to augment A with the vector above to make the next part of this question easier.

NOTE: Maximal partial pivoting will not be tested on the actual test explicitly, but you should understand it!

(b) Suppose that $A\vec{x} = [12, 1, -30, 6]^T$. Find $\vec{x}$.

2. (5 points) Consider the matrix $B = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 7 & 1 \\ 2 & 4 & 1 \end{pmatrix}$. Is B invertible? If so, find its inverse. If not, explain why.

3. (9 points) Which of the following is a vector subspace of $\mathbb{R}^3$? Justify your answers. Answers with no justification will receive no credit.

   (a) All vectors $[b_1, b_2, b_3]^T$ such that $b_1 + b_2 - 2b_3 = 0$.

   (b) All vectors $[b_1, b_2, b_3]^T$ such that $b_1 \leq b_2 \leq b_3$.

   (c) All linear combinations of $[1, 2, 4]^T$ and $[2, 3, 4]^T$.

4. (6 points)

   (a) Suppose that $\vec{v} = [1, 1]^T$ and $\vec{w} = [1, 5]^T$. Find a number c so that $\vec{w} - c\vec{v}$ is perpendicular to $\vec{v}$.

   (b) Suppose that $\vec{v}$ and $\vec{w}$ are both non-zero vectors. Find a formula for a number c so that $\vec{w} - c\vec{v}$ is perpendicular to $\vec{v}$.

5. (5 points)

(a) Briefly explain why for any two matrices E and F, the column space of EF is contained in the column space of E.

(b) Find a matrix F with no zero entries such that the column space of EF is not equal to the column space of E. You do not need to compute the column spaces of either matrix in order to answer this question, but you can use the following matrix if you wish:

$$E = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & 4 \\ 0 & 0 & 7 \end{pmatrix}.$$